MuJoCo 3.17 "flex" for our bonded PDMS skin: how it computes, how it compares with exact solutions, which cell size to use, and how to run it on the robot

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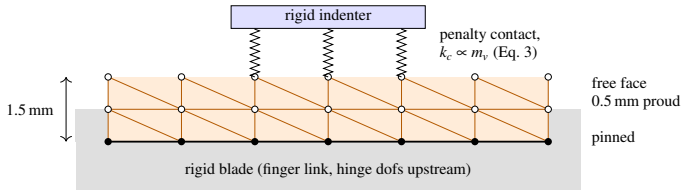


Figure 1: Cross-section of the model. Filled dots: back-face vertices pinned to the blade body; open dots: vertex bodies with 3 slide joints each. Every hex is split into 6 tets along one fixed diagonal. Contact acts as a mass-scaled penalty spring between the indenter and the layer.

Setting. Each blade of the 2F-140 carries a PDMS layer $51 \times 71 \times 1.5$ mm (x across, y along the finger, z through), bonded at its back face, free at the front, read out by 5×7 taxels (10.2×10.14 mm). Material: linear elastic, $E = 1$ MPa, $\nu = 0.45$ (placeholder; also 0.49 and, for the ground truth, 0.499), $\rho = 965\,\mathrm{kg/m^3}$, no damping of any kind. Stiffnesses are in N/µm. Benchmarks: P0 patch tests; G1 laterally confined layer; C1/C1f rigid platen over the whole face, no-slip/frictionless; C2 same, shear along y; C3 platen over $y \geq 5.07$ mm; C4 flat circular punch $a = 5$ mm; C5 sphere $R = 10$ mm to 30 µm; C6/C6f square punch exactly over the centre taxel.

Bottom line. (i) MuJoCo's tetrahedral flex (`dof="full"`) uses a Lamé λ twice too large; a two-line input remap fixes it exactly. (ii) With the remap the elastic operator converges to the exact solution from above at first order in the cell size: +2.6 % on a whole-face load and +36 % on a single-taxel punch at 1.3 mm cells. (iii) Contact is a penalty spring in series with the layer whose stiffness (in N/m) is proportional to the vertex mass, so thinner elements make it softer; on the robot it is the dominant error. (iv) The divergence on the robot is a separate MuJoCo defect (the per-island solver omits the flex stiffness); `<flag island="disable"/>` cures it. (v) Contact is defined by where surfaces touch, not per taxel; in the exact solution the neighbours of a pressed taxel bulge *outward* slightly.

1 What a flex is and how MuJoCo 3.17 computes it

All code pointers are to the 3.17.0 tag (`src/`); they were re-checked for this report.

Data model. `<flexcomp type="grid" dim="3" count="11 15 3">` creates a vertex grid and splits every hex into 6 tetrahedra with the *same* body diagonal in every hex (`user/user_mesh.cc:2703-2738`). With `dof="full"` each unpinned vertex becomes a body with three slide joints (`user/user_flexcomp.cc:529-540`); a pinned vertex (`<pin gridrange>`, our back face) is welded to the parent body (286–379). Every free vertex gets `mass`/(number of points) and pinned vertices' share is dropped (476–477), so our flex carries 2/3 of the specified mass. With `dof="trilinear"/"quadratic"` the degrees of freedom live on a coarse node grid (`cellcount`) and the collision vertices follow by interpolation (580–706) [2]. Vertex joints inherit the parent's `childclass` armature, stiffness and frictionloss (all 0 in our model; keep it so).

Elasticity, `dof="full"` (Saint Venant–Kirchhoff on edges). For a tet of rest volume V with edge lengths l_e (rest L_e), $s_e = l_e^2 - L_e^2$ and the Green strain is $E = \frac{1}{2}\sum_e s_e B_e$, where B_e is built from the two face normals opposite edge e (`ComputeBasis`, 1278–1310). At compile time `ComputeStiffness` (1313–1342) and `MetricTensor` (1179–1220) form

$$k_{ef} = \mu V \,\mathrm{tr}(B_e B_f) + \lambda V \,\mathrm{tr}B_e \,\mathrm{tr}B_f, \tag{1}$$

and the energy evaluated at run time is $s^\top k s/4$ (comment at `engine_core_util.h:323`; forces: `engine_passive.c:587-640`). Substituting E:

$$W = V\left[\mu\,\mathrm{tr}E^2 + \lambda(\mathrm{tr}E)^2\right] \quad \text{vs.} \quad W_{\mathrm{StVK}} = V\left[\mu\,\mathrm{tr}E^2 + \tfrac{\lambda}{2}(\mathrm{tr}E)^2\right], \tag{2}$$

i.e. λ is doubled (finding F1, Sec. 2.2). The energy depends only on edge lengths, so a mirrored (inverted) tet stores no energy: there is no inversion barrier. The experimental Stable Neo-Hookean option (`mjsFlex.elastic3d=1`, MjSpec only, changelog 3.15.0 item 5 [1]; `ComputeSNH` 1345–1372) has the correct small-strain limit.

Elasticity, interpolated (`trilinear`/`quadratic`). Standard Galerkin linear elasticity, $K = \int(\lambda\,\nabla\cdot u\,\nabla\cdot v + 2\mu\,\varepsilon(u):\varepsilon(v))$, fully integrated (`ComputeLinearStiffness`, 1683–1768), co-rotational. Exact in the patch tests. (The `dim=2` membrane path is 4× too soft in shear relative to its documented thickness; not used here.)

Contact. A *plane* geom collides with each flex vertex (`engine_collision_flex.c:216-249`). Any other geom (box, sphere, capsule, mesh) gets one GJK contact per surface-layer tet (`mjc_ConvexElem`, 388): the contact point is a tet corner that jumps between corners from step to step (median 7.2 mm for a flat platen), and its force is spread over the four corners by inverse distance (`engine_core_util.c:618-657, 785-830`), so on average 21 % lands on vertices below the free face (up to 45 %). Contacts are soft constraints: for a contact on one vertex of mass m_v the regulariser is $R = \frac{1-d}{d}\,\mathrm{diag}A$ with $\mathrm{diag}A = 1/m_v$. The resulting static spring per vertex contact (measured to 10^{-6}) is

$$k_c = \frac{d^2}{1-d}\,\frac{m_v}{d_{\max}^2\,\tau^2\zeta^2} \;\leq\; \frac{0.9999}{0.0001}\,\frac{m_v}{h^2}, \tag{3}$$

(τ, ζ from `solref`, d from `solimp`; the ceiling is the discrete-integrator floor on R, `engine_core_constraint.c:2119-2121`). With the robot's `solref="1e-4 1"`, $d = 0.99$ and $m_v = 10.6$ mg ($11\times15\times3$ grid) $k_c = 1.06 \times 10^5$ N/m per vertex, 17.5 N/µm over the free face: only twice the layer's own 8.57 N/µm. The ceiling is 69.6 N/µm for the face at any `solref`. Contacts of rigid and interpolated flexes are reduced to 50 per geom (`flexReducible`, `engine_collision_driver.c:572-575`; changelog 3.16.0 item 10).

Discrete integrator. Flex elasticity requires `integrator="discrete"` (implicit/implicitfast refuse it). Each step solves for the velocity in the metric $\hat{M} = M + hD + h^2K$ by PCG (`mj_effSolve`, `engine_metric.c:1638-1714`) with 3×3 vertex-block preconditioning when the flex is fixed to a static body, and matrix-free with a mass preconditioner when its parent has joints.

Table 1: Ground truth (N/μm; C5 force in N, contact radius a in mm). Errors: C1–C3 $\leq 0.05\,\%$, C4–C6 $\leq 10^{-4}$. $GA/t = 0.8324$ at $\nu = 0.45$. MA/t (Winkler line, comparison only) is +6.9, +22, +108 % above C1.

case	$\nu = 0.45$	0.49	0.499
G1 $= MA/t$	9.1566	41.313	403.41
C1 platen, no-slip	8.5662	33.757	194.33
C1f platen, frictionless	8.0102	27.252	91.727
C2 platen shear along y	0.82791	0.80602	0.80126
C3 half platen	3.5927	13.139	55.376
C4 punch $a = 5$, frictionless	0.14757	0.24139	0.28798
C6 taxel punch, no-slip	0.21658	0.48158	0.73233
C6f taxel punch, frictionless	0.19971	0.34327	0.42106
C5 $R = 10$ mm, $F(30\,\mu\text{m})$ [N]	0.04397	0.04822	0.04938
C5 contact radius a [mm]	0.632	0.640	0.642

The PCG budget is `opt.iterations`; exhausting it raises the "too ill-conditioned" warning, and a non-positive curvature exits silently (1679–1681). A flex on a jointed parent (supported since 3.15 [4]) requires `solver="CG"` (`engine_forward.c:1753`). Our layer has $h\omega_{\max} = 22$–916 at $h = 0.5$ ms and first thickness modes at 10.45 kHz (compression) and 3.15 kHz (shear), $\omega h = 32.8$ and 9.9; the integrator is accurate only for $\omega h \leq 0.3$ [6]. The elastic part is unconditionally stable, but the layer's dynamics are numerically over-damped: *treat the flex as quasi-static.*

2 Verification against exact solutions

2.1 Ground truths (no MuJoCo involved)

Closed forms. P0: $\sigma_{zz} = M\varepsilon$, $\tau = \mu\gamma$, $p = 3K\varepsilon$. G1: $K = MA/t$, $M = \lambda + 2\mu$, which is also a rigorous *upper* bound for C1 for any conforming discretisation. **Infinite bonded layer** (`gt_transform`): Fourier/Hankel transform of the Navier equations solved with Papkovich–Neuber potentials, $u = 0$ at $z = t$; checked symbolically, against an independent Chebyshev ODE solve ($\leq 2.7 \times 10^{-6}$), against both limits (t/M, half-space), and against the published kernel of [7]. Contact problems (C4, C5, C6) solved as integral equations (boundary elements, Richardson extrapolation) (C6 error $\leq 9 \times 10^{-5}$). **Finite pad** (`gt_fem`): 27-node quadratic bricks, graded to 0.1 mm at edges, three nested levels (up to 3.4 M dof), mixed Q2/P1 form at $\nu = 0.499$; verified by patch tests (6×10^{-13}), G1 (10^{-14}), manufactured solutions (rates 3.17/2.06 vs. 3/2) and a no-locking test of the mixed form. The two ground truths agree within their error bars where they overlap (C4 to 2.7×10^{-5}, C6 to 1.2×10^{-3}). C1–C3 are taken from the FEM, C4–C6 from the transform solution (Table 1). For comparison, the classical thin-layer pressure method [8] gives C1 = 7.337 N/μm ($-14.3\,\%$) at $\nu = 0.45$.

Two physical facts follow. Going from $\nu = 0.45$ to the realistic 0.49–0.499 raises flat-contact stiffness 2–3.4× (shear is unchanged), so the placeholder matters. And the Winkler foundation ($k = M/t$) is not a usable reference: +35 % on C4 at $\nu = 0.45$, up to 30× at 0.499, and the wrong shape for the sphere.

2.2 Patch tests and the doubled λ (F1)

Exact homogeneous strains were imposed and `qfrc_spring` compared with the exact nodal forces (Table 2). Fitting $f = \mu f_\mu + \lambda f_\lambda$ gives $\lambda_{\text{eff}}/\lambda = 2.00000$ (residual $< 10^{-11}$) and $\mu_{\text{eff}}/\mu = 1.00000$ for $\nu = 0.3$, 0.45, 0.49, on a cube and on our layer, with rotated and moving parents, and at 1 % finite strain: Eq. (2) exactly. The simulated material at $\nu = 0.45$ is $\nu_{\text{eff}} = 0.4737$ with E almost unchanged (×1.016) but bulk modulus ×1.931 and confined modulus ×1.818. This is why nobody noticed: MuJoCo's own test uses $\nu = 0$ (`test/engine/engine_passive_test.cc:578`) [3], and the validations we found are cantilevers or unconfined blocks; a thin bonded layer is the worst case. No upstream report exists (searched in the survey; PR #3621 only notes that "the previous 3D StVK volumetric normalization is preserved" [5]).

Table 2: P0 patch tests: energy ratio MuJoCo/exact over 154 models (from `bench_static`).

variant	uniaxial strain	simple shear	hydrostatic
full, raw inputs, $\nu = 0.45$	1.818182	1.000000	1.931034
full, raw inputs, $\nu = 0.49$	1.960784	1.000000	1.986577
full remapped; trilinear; quadratic	1.000000	1.000000	1.000000

Table 3: Discretisation error of MuJoCo's elastic operator (exact BCs, no contact), $\nu = 0.45$, %. Meshes in cells $n_x \times n_y \times n_z$ (vertices = cells +1); the robot grid 11×15×3 vertices is 10×14×2 cells (5.1 mm).

mesh	C1	C1f	C2	C3	C6
10×14×2, raw (robot now)	+85.3	+77.3	+0.5	+100.4	+345
10×14×2, F1-corrected	+4.4	+4.8	+0.5	+14.2	+158
5×7×1, corrected	+6.9	+9.2	+0.5	+26.1	+371
40×56×2, corrected (1.275 mm)	+2.6	+2.3	+0.5	+5.1	+36.3
quadratic 40×56×3	+0.3	+0.3	+0.2	+0.7	+5.1
40×56×2 corrected, $\nu = 0.49$	+7.8			+14.1	+110

Workaround (exact, verified): pass $\nu_{\text{in}} = \nu/(2(1-\nu))$, $E_{\text{in}} = E(1+\nu_{\text{in}})/(1+\nu)$, which keeps μ and feeds $\lambda/2$: $\nu = 0.45 \Rightarrow$ `young="971787" poisson="0.409091"`; $\nu = 0.49 \Rightarrow$ `993552`, `0.480392`. **Upstream fix:** in `ComputeStiffness` (`user_flexcomp.cc:1325`) divide `la` by 8 instead of 4 (not in `MetricTensor`, which the SNH path reuses with `la`$= -\mu$). A report draft with a 17-line repro (one tet; prints ratios 1.8182 and 2.0000) is in `study/internals/bug_lambda.md`; it has not been posted.

2.3 The elastic operator vs. ground truth

To separate discretisation from contact, MuJoCo's own stiffness was extracted by finite differences of `qfrc_spring` (symmetric to 7×10^{-10}, equal to MuJoCo's assembled matrix to 1.6×10^{-9}) and loaded with exact displacement boundary conditions on 295 meshes. Table 3 gives the errors. With F1 uncorrected the current robot grid is +85 % on C1; F1 is the dominant error for distributed loads. Corrected, every element type converges from above, consistent with the MA/t bound. Shear (C2) is within 0.55 % on every mesh. At $\nu = 0.499$ all linear elements lock (corrected full flex with 6 elements through the thickness is still +5.5 % on C1). The single fixed tet diagonal also biases local loads: a normal load on the centre taxel produces a spurious in-plane displacement of 16 % of u_z on the 11×15×3 grid (6 % at 21×29×5), which a mirrored split removes.

3 Cell size: accuracy, cost, convergence

In-plane size (Fig. 2). Convergence is first order for every element type. For distributed loads (C1, C3) refinement below ≈2.5 mm buys little: with contact, C1 moves from −12.4 to −13.7 % between 2.55 and 1.275 mm while cost rises 4.4× and the contact count 3.8×. For localized loads the error falls only as $\Delta x^{1.2}$: C6 is +72, +36, +15 % at 2.55, 1.275, 0.64 mm. The C5 sphere's exact contact radius (0.63 mm) is smaller than any affordable uniform cell, so C5 is +313 % at 1.275 mm.

Through the thickness (Fig. 3a). The static error saturates after 2–3 elements (C1 +3.2, +2.5, +2.3 % for 2, 3, 4 at 2.55 mm). With contact, more elements make the reading *worse*: by Eq. (3) the contact stiffness scales with the vertex mass, and thinner elements mean lighter surface vertices. The contact error is −15,

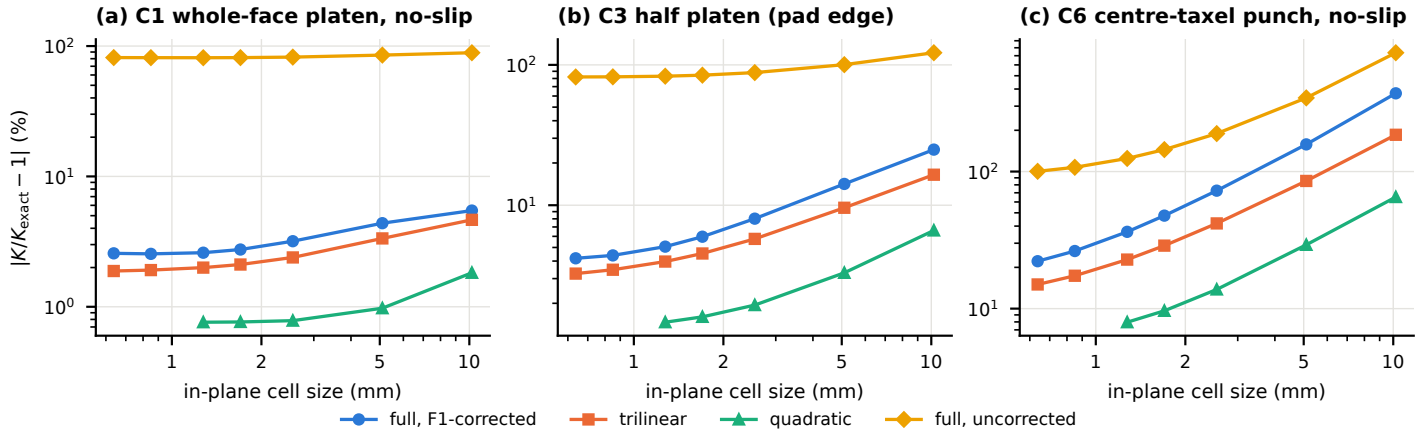


Figure 2: Discretisation error vs. in-plane cell size (2 elements through the thickness; quadratic: 1 cell), $\nu = 0.45$. Fitting $e = a\Delta x^p + b\Delta z^q$ gives $p = 0.76, 0.97, 1.22$ for C1, C3, C6 (full, corrected): first order, as the edge singularities allow; quadratic elements lower the constant, not the order. The raw (uncorrected) flex has an ≈80 % floor on distributed loads.

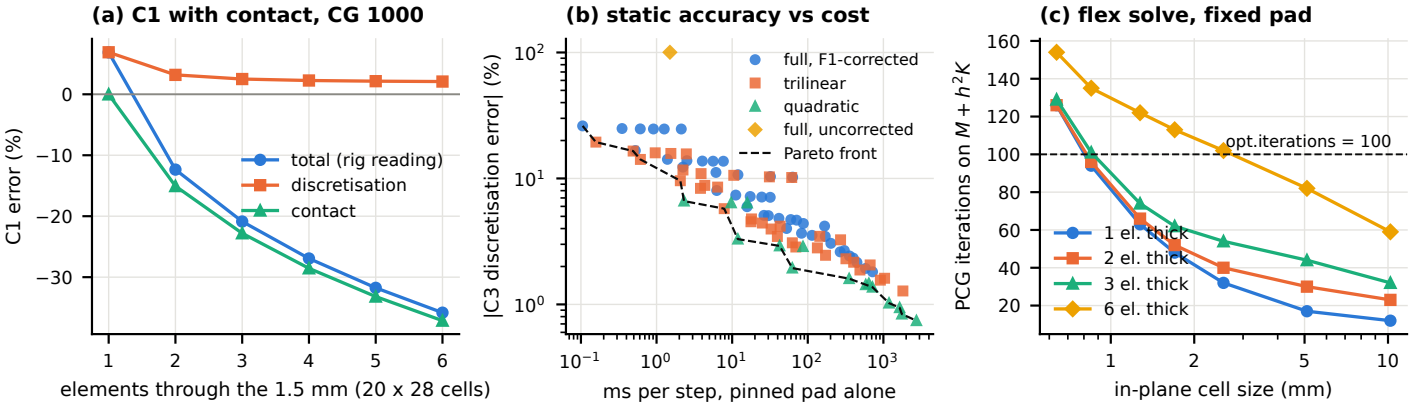


Figure 3: (a) Whole-face platen with contact (robot contact setting, 20×28 cells): the static error saturates, but each extra element through the thickness makes the contact (and the total reading) softer. (b) Static accuracy against time per step for all 295 meshes (shared machine; indicative). (c) PCG iterations of the flex solve ($M + h^2K$, fixed pad) against in-plane cell size; above 100 the default budget is exceeded.

−23, −29, −33, −37 % for 2–6 elements; the total goes from −12 to −36 %. Each added element also costs 1.3–1.4× and adds PCG iterations (40, 54, 70, 86, 102 at 2.55 mm for 2–6). This is the **overmeshing** to avoid: through-thickness refinement beyond 2 elements, and in-plane refinement below 2.5 mm for distributed loads.

Solver convergence. The stiffness of the discretised layer makes $\mathrm{cond}(M + h^2K)$ range from 34 to 6.1×10^4, set mainly by the element thickness. The flex PCG needs 12–419 iterations and exceeds the default 100 at ≈0.85 mm cells (Fig. 3c). The contact solve is the harder part: with the robot's elliptic cone and `impratio=100`, no-slip contact never converges in 100 CG iterations on any mesh (every step hits the cap; force chatter 3×10^{-2} to 2×10^{-1}, and the C2 shear force flips sign). With 1000 iterations the median is 480–800 and the force settles within ≈1 ms on 2.55 mm or finer meshes. Frictionless contact converges in 15–70 iterations.

Recommended cell structure. *Robot* (budget 1–5 ms per step): 6×8×2 vertices (5×7×1 cells), one element per taxel with grid lines on the taxel borders and one through the thickness, F1-corrected. *Fixed rig* (sensor studies): 41×57×3 vertices (40×56×2 cells, $1.275 \times 1.268 \times 0.75$ mm), and for an indenter at a known position a graded tet mesh (`flexcomp type="direct"`): with a 0.5 mm core C4, C6, C6f are $+14.8 \pm 7.3$, $+16.6 \pm 10.0$, $+17.8 \pm 7.8$ %, and with a 0.15 mm core C5 is +11.6 %, at 0.1–0.5 s per step and with only 3–5 % of the indentation taken by contact penetration. Table 4 gives the total errors (contact included) of both choices. Interpolated flexes, though most accurate per dof statically, are rejected with contact: the 50-contact cap makes their apparent stiffness collapse with refinement (trilinear C1f 1.82, 0.36, 0.10 N/µm at 10×14×2, 20×28×3, 40×56×3) and no-slip shear fails (−93 to −133 %).

Table 4: Total error (%, contact included, $\nu = 0.45$) of the recommended structures on the rig; ± is the 95 % interval over 5 indenter placements. Rig: `solref 1e-4`, CG 1000. Robot: `solref 1e-3`, CG 100.

case	rig 40×56×2	(discretis. / contact)	robot 5×7×1
C1	−13.7	+2.6 / −15.9	−89.0
C1f	−12.6	+2.3 / −14.6	−88.3
C2	−8.2	+0.5 / −8.7	−41.1
C3	−12.4	+5.1 / −16.7	−88.4
C4	+8.4 ± 7.0	−1.8 / +10.4	−70.7 ± 13.9
C6	+9.1 ± 24.8	+36.3 / −20.0	−76.4 ± 5.5

On the rig, the good C4/C6 numbers of the uniform mesh are a *cancellation* (+36 % discretisation, −20 % contact) with a ±25 % dependence on where the punch sits relative to the grid; the graded mesh's errors are genuine. At $\nu = 0.49$ the layer is 4× stiffer but the contact is not, so the contact error doubles (rig C1 −46.9 %). For deformation studies read the layer from the vertex positions (`flexvert_xpos`), not from the indenter travel.

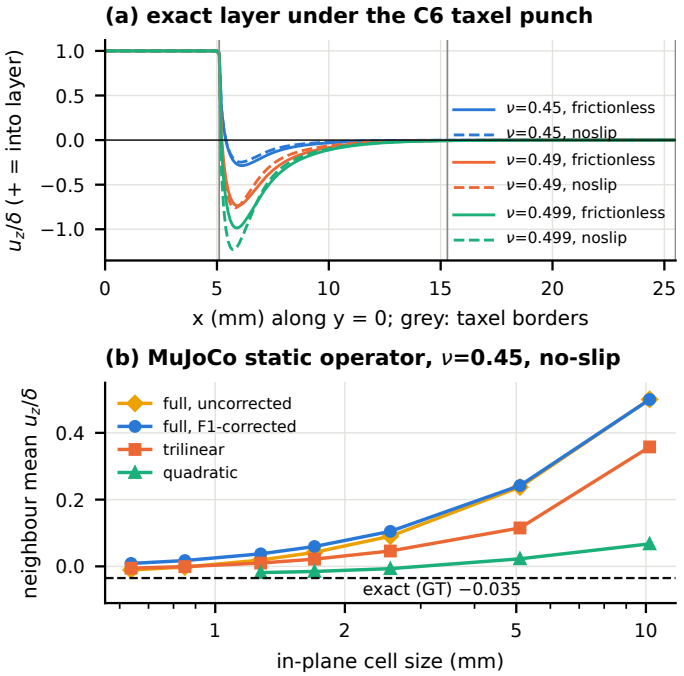


Figure 4: (a) Exact surface displacement along $y = 0$ under the C6 taxel punch (gt_transform). Outside the punch the surface rises; the neighbours carry no contact force. (b) Mean u_z/δ of the four neighbouring taxels in MuJoCo's static operator vs. cell size; positive values on coarse meshes are interpolation spill-over through shared nodes.

4 Contact with objects: what the layer does

Is contact defined per cell? No. In MuJoCo, as in the continuum, contact exists where the surfaces touch; `tactile.py` (`PDMSSkin`) bins each contact force into the taxel that contains its contact point. In the exact solution for the C6 punch the contact force is 100 % in the centre taxel and exactly zero in the neighbours, and the neighbouring surfaces do not follow the punch: they bulge outward (Fig. 4a). The mean u_z/δ of the edge neighbours is $-0.047/-0.035$ (frictionless/no-slip) at $\nu = 0.45$, $-0.147/-0.127$ at 0.49 and $-0.200/-0.202$ at 0.499; the peak rise just outside the punch edge is 0.29, 0.74 and 0.99δ. So *if cells 1, 2, 3 deform* $+1, +2, +1$*, the outer cells must themselves be touched* (and carry force), or the pattern is a mesh artefact: pressing cell 2 alone by 2 units moves cells 1 and 3 by about -0.07. What does spread is the force transmitted to the rigid blade: at the bond plane the centre cell takes 84.2/88.5 % (0.45) and each edge neighbour 1.6–3.6 %.

MuJoCo's static operator reproduces this only on fine meshes (Fig. 4b): on the robot grid the neighbours move $+0.22$ to $+0.25\delta$ (and unequally, 0.252 vs. 0.223, because of the fixed tet diagonal); quadratic elements at 1.275 mm give -0.024. With contact, the force still bins only into touched taxels; neighbours get displacement only (27–35 % of the centre at $10\times14\times2$, 0.3–7 % at $40\times56\times3$).

Close-up video (Fig. 5). Eight rigid objects (sphere, rod, box edge, book corner, capsule, ellipsoid, coin, cube) are pressed into the fixed pad by servo-driven slide joints, slid and lifted. All runs are free of NaN and inverted tets (minimum volume ratio 0.754), CG never hits its cap, the taxel forces sum to the total contact force to 5×10^{-15} N, and the residual displacement after lift-off is $\leq 1.2\times10^{-4}$ µm. The displacement is one continuous field across taxel borders while force goes only to touched taxels; for the cube: taxel (3,3) 3.18 N with mean $u_z = 14.1$ µm, (3,4) 0.61 N and 3.8 µm, (3,2) 0 N and -0.1 µm. Limitations, stated plainly: (i) a 1 MPa, 1.5 mm layer indented by tens of micrometres carries only ≈1 N under a curved object; seven of eight objects stopped at the 100 µm indentation cap at 0.7–4.2 N, not the 5–20 N targets. A 250 µm cap inverted 6 surface tets during sliding, which stayed inverted after lift-off (no inversion barrier). (ii) Even with the stiffest contact, penetration is 11–30 % of u_z. (iii) Contacts smaller than a cell are under-resolved (the sphere dimple is 1–2 vertices wide; the book corner hops between vertices). (iv) With no material damping, the hold rings (cube: 6.6 µm std).

Table 5: 2.7 s close-and-lift with both pads, islands off, F1-corrected; held force L/R, load on the rigid blade, largest contact penetration, wall time per 0.5 ms step (single thread, shared machine). A uses tolerance 10^{-10}.

grid	`solref` / CG its	held L/R (N)	rim (N)	pen. (mm)	ms/step
11×15×3	now: `1e-4`/100	45/28	158	0.03	24.1
11×15×3	A: `1e-4`/1000	120.2/120.1	0	0.055	145
11×15×3	B: `1e-3`/100	121.2/121.9	9.1	0.62	27.7
6×8×2	B (recomm.)	119.7/119.1	0	0.17	6.0
6×8×2	A	120.8 (L)	–	0.01	21.4

5 Running it on the robot

Failure. Pinned to the blade inside the 2F-140, the full flex diverged within ≈3 steps whenever any upstream joint was free. **Cause** (found in the source, reproduced in a minimal model, fixed in an instrumented build): the constraint solver works per *island*, and the island metric operator selects flexes with `flexMatrixFree` (`engine_metric.c:2485-2491`), which gates on `mjd_flexStiff_active` (`engine_derivative.c:1901-1908`). That returns 0 for any flex that is not "simple", i.e. pinned to a body with joints. The monolithic operator `mj_effMulAdd` (`engine_metric.c:236-246`) does include such flexes, so the smooth acceleration is right but the constraint correction is computed with M alone. The largest generalised eigenvalue of $(M+S_{\text{isl}})^{-1}(M+S_{\text{mono}})$ is 1.5×10^4 on the robot (1.000 with the fix), equivalent to explicit integration at $h\omega \approx 120$. Any constraint row creates an island (mimic equalities, a joint limit, a contact), which is why locking all joints was stable. The PCG budget is *not* the cause (the home pose needs 116 iterations against 100, hence the warning; the solve never takes the $pAp \leq 0$ exit). Minimal repro: one hinge, a pinned $3\times3\times2$ flex and one joint equality diverge at step 3 (`study/robot/repro/minimal_repro.xml`). **Fix:** `<flag island="disable"/>`, or the one-function patch `study/robot/fix/island_flex_fix.patch` (same results).

Second problem: the contact setting. With islands off, the robot's `solref="1e-4 1"` (time constant $h/5$) and CG 100 overload the layer (≈1000 N per side within 25 ms against a 120 N squeeze); it turns inside out (798/866 inverted tets, free face 1.6–3.0 mm in) and the box ends up held by the rigid rims. Two settings work (Table 5): A keeps the stiff contact and lets CG converge; B softens the contact to $\tau = 2h$.

Setting B is cheap and its forces are steady, but the contact supplies most of the compliance: on $11\times15\times3$ its 0.62 mm penetration exceeds the 0.5 mm the pad stands proud, and on the rig the recommended $6\times8\times2$ version reads C1 −89 % (Table 4). Taxel *forces* follow equilibrium (force-map RMS 1.6–3.6 % of the total) and stay in the pressed taxel; taxel *deformations* and the skin's stiffness are wrong. If those matter on the robot, use A (21 ms/step on $6\times8\times2$). Stiffening B through `solimp` 0.999/0.9999 or a contact time constant below $2h$ brings the overload-and-invert failure back; scaling the flex mass makes the compression chatter and breaks

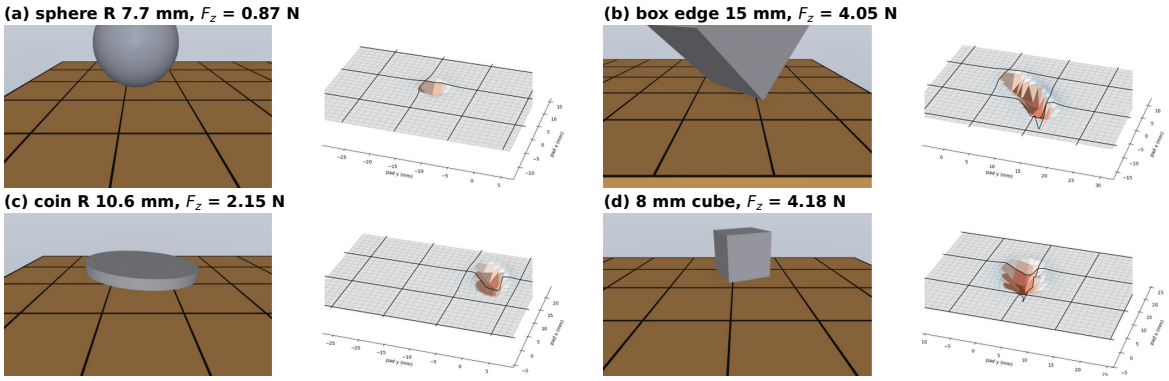


Figure 5: Frames from the close-up video at full normal load, before sliding (each pair: true-scale MuJoCo render with taxel borders; free-face displacement exaggerated 40×, colour = u_z in µm). 40×56×3 cells, F1-corrected, `solref 3e-6`, `solimp 0.9999`, CG 3000. Video: `docs/media/flex_contact_closeup.mp4` (1920×1080, 41.7 s, 8 objects: approach, press, hold, slide 2.1–4.0 mm, hold, lift-off).

Table 6: Settings for the PDMS layer in MuJoCo 3.17.0.

item	setting
element	`flexcomp type="grid" dim="3" dof="full"`, radius 0, back face `<pin gridrange>`
material	`young="971787" poisson="0.409091"` for $E = 1$ MPa, $\nu = 0.45$ (`993552/0.480392` for 0.49); damping 0
grid, robot	6×8×2 vertices (one cell per taxel, 1 thick)
grid, rig	41×57×3; graded `type="direct"` mesh for local indenters
mass	ρV; note the flex carries $(n - n_{\text{pin}})/n$ of it
contact	`priority 2`, `friction 1`, `solimp "0.99 0.99 0.001"`; `solref "1e-3 1"` robot (B), `"1e-4 1"` rig/setting A
option	`integrator="discrete" solver="CG"`, $h = 0.5$ ms, `cone="elliptic" impratio="100"`, `refsafe` disabled, `<flag island="disable"/>`
iterations	100 (robot B); 1000, tol 10^{-10} (rig, setting A, any no-slip study)
read-out	forces: per contact point, binned (`tactile.py`); deformation: `flexvert_xpos`; log inverted tets

the density rule. Interpolated flexes are unsuitable on the robot for a further reason: their node forces are written only to the node's own dofs and skip ancestor joints (`engine_passive.c:160-173`), so an upstream slide joint feels +2.53 N (trilinear) where momentum balance requires 0.

The full librarian task does not work yet. With setting A on 6×8×2 the task ran without divergence, but the book slipped 45 mm and turned 31° in 0.2 s when it left the shelf: the skins had reported $q \approx 0.7$–0.8 during the pull (cells 0.53), so the controller squeezed 23 instead of 28 N per blade. (i) In a torsion test (`study/torsion/`) the box turns at 0.58 N m on the 6×8×2 flex against 1.11 N m on the cells at 24 N per pad (2.21 vs 4.61 at 121 N); the exact pressure (radius 22.7 mm) gives $\mu \sum N\bar{r} =$ 1.11 N m, the cells' value. Box–tet contacts sit at tet corners and pull the force toward the pad centre, halving the torsional grip. Finer pads do not cure it: 11×15×2 gives 0.49 N m (contacts reach ≈30 mm from the centre, cells 38–40 mm), and 11×15×3 spreads them to 38 mm but makes them softer, 0.31 N m (the layer's own twist would need 18 N m per degree). (ii) `mj_rnePostConstraint` skips flex contacts (`engine_core_smooth.c:2531`), so `cfrc_ext` and force/torque sensors ignore loads on the pads (wrist F/T 0.03 N during the pull vs 3.2 N with cells).

6 Recipe, open issues, verified vs. assumed

Open issues. (1) Three engine defects are documented but not reported upstream: the doubled λ, the island metric omission, and the interpolated-flex force path; also the false compile warning "not rigid and has no equality constraints or passive forces", which ignores `young` (`user/user_model.cc:3694-3702`). (2) The SNH option (`elastic3d=1`), exact in the patch tests, has not been tried on C1–C6 or on the robot. (3) Contact compliance scales with vertex mass and caps at $\approx 10^4 m_v/h^2$; at realistic $\nu \geq 0.49$ the robot skin is almost all contact spring. (4) The fixed tet diagonal causes shear cross-talk; a mirrored `direct` mesh removes it but was not combined with grading. (5) Dynamics of the layer are not physical at $h = 0.5$ ms; no viscoelasticity was modelled (by design).

Verified vs. assumed. Verified by runs: every number above (scripts and outputs in `docs/report_flex/study/{internals, gt_transform, gt_fem, bench_static, bench_dynamic, compare, robot, video}`), and every code pointer by reading the 3.17.0 source. Not verified: the upstream fix for λ was not built (it is algebraically identical to the verified input remap); the island patch was tested only on our dim-3 models; timings were measured on a shared machine; the explanation of why graded meshes get stiff contacts (uniform vertex mass) is an inference; the contact ceiling rests on three `solref` values. Corrections to the earlier findings: the 50-contact cap is changelog 3.16.0 (not 3.17); F2's platen values came from a welded rig, and its 18.9 N/µm exceeds the 16.65 N/µm that a doubled-λ discretisation can reach; F4's divergence is the island defect, not ill-conditioning or the $pAp \leq 0$ exit.

References

[1] MuJoCo 3.17.0, *Changelog* (`doc/changelog.rst`), `https://github.com/google-deepmind/mujoco/blob/3.17.0/doc/changelog.rst`.

[2] MuJoCo 3.17.0, *XML reference* (flex, flexcomp, option), `https://github.com/google-deepmind/mujoco/blob/3.17.0/doc/XMLreference.rst`.

[3] MuJoCo 3.17.0, `test/engine/engine_passive_test.cc`, `https://github.com/google-deepmind/mujoco/blob/3.17.0/test/engine/engine_passive_test.cc`.

[4] MuJoCo pull request #3618, flex elasticity through vertex attachment Jacobians, `https://github.com/google-deepmind/mujoco/pull/3618`.

[5] MuJoCo pull request #3621, Stable Neo-Hookean elasticity via `mjsFlex.elastic3d`, `https://github.com/google-deepmind/mujoco/pull/3621`.

[6] cps-test-lab, *roqsim*, `flex_modes.py` (measured discrete-integrator damping and $\omega h \leq 0.3$ rule), `https://github.com/cps-test-lab/roqsim`.

[7] I. Argatov, A. U. Daniels, G. Mishuris, S. Ronken, D. Wirz, "Accounting for the thickness effect in dynamic spherical indentation of a viscoelastic layer: application to non-destructive testing of articular cartilage," arXiv:1203.0918, 2012.

[8] J. M. Kelly, N. C. Van Engelen, "Single series solution for the rectangular fiber-reinforced elastomeric isolator compression modulus," PEER Report 2015/03, UC Berkeley, 2015.